Core practical – Investigating inks, Part 1

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

Name _____ **Class** _____ **Date** _____

Part 1: Distillation procedure

Practical Objective

To use distillation to produce pure water from ink.

Apparatus

- eye protection
- conical flask with side arm
- delivery tube
- test tube
- ink
- Bunsen burner
- gauze
- rubber bung with thermometer
- 250 cm^3 beaker
- ice
- tripod
- heat-resistant mat
- anti-bumping granules

Safety ⚠

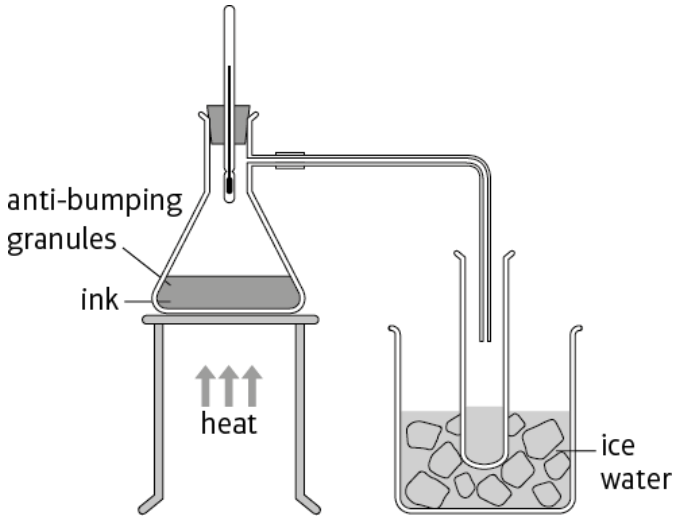
- Eye protection should be worn at all times.
- Anti-bumping granules should be used to reduce the risk of the liquid boiling over

Your teacher may watch to see if you can:

- carry out experiments safely, reducing the risks from hazards.

Method

A Set up your apparatus as shown in the diagram. Put anti-bumping granules in the bottom of the flask.



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B Adjust the Bunsen burner so that you have a gentle blue flame. The air hole should be about half open and the gas tap should be about half on.

C Heat the ink until it boils.

D Collect the distillate in the test tube and note the temperature of the vapour.

Planning and predicting

1 When you distil the ink, how will you know if you have successfully purified the water?

_____

_____

_____

2 Predict the temperature reading on the thermometer when the ink is boiling. Explain your answer.

_____

_____

3 What is the purpose of the ice water shown in the diagram?

_____

_____

4 The conical flask might be knocked off the tripod.

a Why is the conical flask a hazard if knocked over?

_____

_____

b How can the risk of harm from this hazard be reduced?

_____

_____

5 Suggest one other hazard and a way of reducing the risk from this hazard.

_____

_____

6 What air hole and gas settings should you have for the Bunsen burner:

a when you are not using it

_____

_____

b when you are using it to heat the ink?

_____

_____

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Considering your results/conclusions

7 Did you purify the water successfully? Explain your answer. Try to include a possible test you could carry out to show it was water.

_____

_____

_____

8 Explain what happened when the ink was distilled. In your explanation, use the following words: boil, evaporate, liquid, steam, temperature, vapour.

_____

_____

_____

_____

_____

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Core practical – Investigating inks, Part 2

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

Name _____ **Class** _____ **Date** _____

Part 2: Chromatography procedure

Practical Objective

You are going to test some inks to see how many dyes they contain and calculate their R_f values.

Apparatus

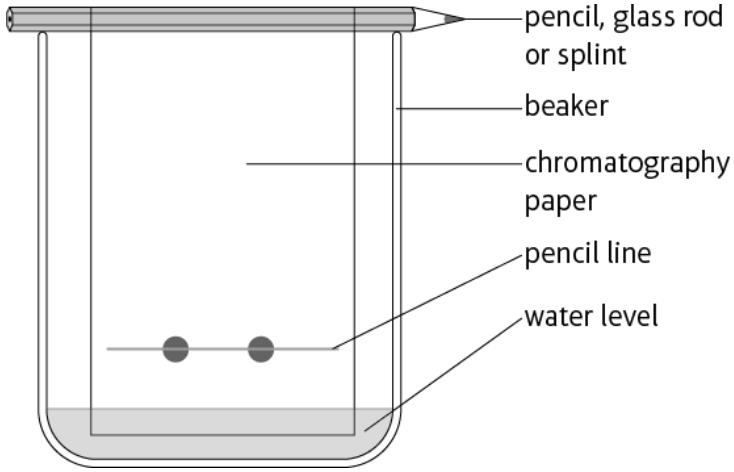
- pencil and ruler
- beaker
- chromatography paper attached to a pencil, rod or splint
- two marker pens or felt-tip pens

Your teacher may watch to see if you can:

- follow instructions carefully.
- draw conclusions from your results.

Method

A Check that your chromatography paper hangs close to the bottom of the empty beaker without touching it (as shown in the diagram).



B Take the paper out of the beaker and draw a pencil line on the paper, about 2 cm from the bottom.

C Put a small spot of ink from each pen on your pencil line.

D Write the name of each pen or ink below each spot with a pencil.

E Pour some water into the beaker to a depth of about 1 cm.

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F Lower the chromatography paper into the beaker so that the bottom of the paper is in the water, but the water level is below the spots (see the diagram).

G Leave the paper in the beaker until the water soaks up the paper and reaches near the top of the paper. The water is the solvent for the different coloured dye compounds in each ink. The solvent is called the mobile phase in chromatography, because it is the part that is moving.

H Take the paper out and immediately use a pencil to mark the location of the solvent front (the level the water has reached) before it evaporates. Leave the paper to dry.

Recording your results

1 Each ink is a mixture of several coloured dye compounds. Describe the coloured dye compounds that mix to produce the ink in each pen.

_____

_____

_____

_____

_____

_____

2 Measure the distance the solvent (the water) has risen from the pencil line.

_____

3 Measure the distance that each dye spot has risen from the pencil line. Measure from the pencil line to the top of each different coloured spot. Write your results in the tables below.

Name of pen/ink				
Colours of dye spots				
Distance of spot from pencil line (cm)				
R_f value				

Name of pen/ink				
Colours of dye spots				
Distance of spot from pencil line (cm)				
R_f value				

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Considering your results/conclusions

4 Calculate the R_f value using the equation below for each separate colour in the inks. Add this value to your table above.

$$R_f = \frac{\text{distance moved by the coloured spot}}{\text{distance moved by solvent}}$$

5 Were any of the inks a pure colour? Explain your conclusion.

6 Did the same coloured dyes appear in more than one ink? If so, do you think they were the same chemical compound? Explain your answer.

Evaluation

7 Why was the starting line drawn in pencil?

8 Why did you have to label the spots?

9 Why is the chromatography paper hung with the bottom just in the water?

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10 Why must the water level in the beaker be below the spots?

11 How easy was it to identify the level to which each coloured dye had travelled? How would this affect the accuracy of the R_f values you calculated?

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